

Question 1

By Simpson's Rule $\int_{-0.1}^{0.1} f(x) dx \approx \frac{h}{3} [f_0 + f_2 + 4f_1]$

where $h = \frac{0.1 - (-0.1)}{2} = 0.1$

$\therefore \int_{-0.1}^{0.1} f(x) dx \approx \frac{1}{30} [-1 + 0.5 + 0 + 4]$
 $= \frac{3.5}{30} \approx \underline{\underline{0.116}}$

Truncation Error: $\frac{h^4(b-a)}{180} |f^{(4)}(r)|$ where $r \in [-0.1, 0.1]$

By MVT, $f^{(4)}(r) - f^{(4)}(0) = f^{(5)}(\xi) \cdot r$, $\xi \in [0, r]$

$f^{(4)}(0) = 0$, $|f^{(5)}(\xi) \cdot r| \leq 0.4$

$\therefore |f^{(4)}(r)|_{\max} = 0.4$

max truncation error $= \frac{0.1^4 \cdot 0.2}{180} \times 0.4 \approx \underline{\underline{4.4 \times 10^{-8} \approx 10^{-7}}}$

Question 2 we have $a=0$, $b=1$. First find n which satisfies truncation error at most 5×10^{-4}

$f(x) = \ln(1+x^2)$ truncation error $= \frac{(b-a)h^4}{180} |f^{(4)}(r)|$, $r \in [0, 1]$

$f'(x) = \frac{2x}{1+x^2}$ $f''(x) = \frac{2(1+x^2) - 4x^2}{(1+x^2)^2} = \frac{2}{(1+x^2)} - \frac{2x^2}{(1+x^2)^2}$

$f^{(4)}(x) = \frac{-2(1+x^2) \cdot 2x \cdot 2}{(1+x^2)^4} - \frac{4x(1+x^2)^2 - 8x^3(1+x^2)}{(1+x^2)^4}$
 $= \frac{8x^3}{(1+x^2)^3} - \frac{12x}{(1+x^2)^3}$

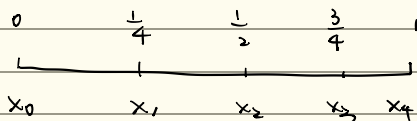
$f^{(4)}(x) = \frac{24x^2(1+x^2)^2 - 32x^4(1+x^2)}{(1+x^2)^4} - \frac{12(1+x^2)^3 - 2x^3(1+x^2)}{(1+x^2)^4}$
 $= -\frac{12}{(1+x^2)^3} + \frac{72x^4}{(1+x^2)^4} + \frac{24x^2}{(1+x^2)^3} - \frac{32x^4}{(1+x^2)^4}$

$f^{(5)}(x) = \frac{36}{(1+x^2)^4} + \frac{216x^2 - 360x^4}{(1+x^2)^5} + \frac{48x - 48x^3}{(1+x^2)^4} - \frac{128x^3 - 64x^5}{(1+x^2)^4} \geq 0$

$\therefore |f^{(4)}(x)|_{\max} = \max\{|f^{(4)}(0)|, |f^{(4)}(1)|\} = 12$

$b-a=1$ $h = \frac{1}{n}$ $\frac{12}{180 n^4} \leq 5 \times 10^{-4}$

$n^4 \geq \frac{12 \times 10^4}{180 \times 5} = 133.33$ $\therefore n_{\min} = 4$



By Simpson's Rule $\int_0^1 f(x) dx$, $f(x) = \ln(1+x^2)$

$\approx \frac{h}{3} [f_0 + f_4 + 2f_2 + 4(f_1 + f_3)]$, $h = \frac{1}{4}$

$f_0 = 0$, $f_1 = \ln \frac{17}{16}$, $f_2 = \ln \frac{5}{4}$, $f_3 = \ln \frac{25}{16}$, $f_4 = \ln 2$

$\therefore \int_0^1 f(x) dx \approx \frac{1}{12} (1^2 + 2 \ln \frac{5}{4} + 4 \ln \frac{17}{16} + 4 \ln \frac{25}{16}) \approx \underline{\underline{0.2639}}$

Question 3

we need first find n which satisfies error $\leq 5 \times 10^{-4}$

Error $= \left| \frac{2}{2n+1} \cdot \left[\frac{2^n (n!)^2}{(2n)!} \int_0^1 \frac{f^{(2n)}(x)}{(2n)!} dx \right] \right|$

By trivial $n=4$

$f(x) = \sin^2\left(\frac{3x}{\pi} + 2\right) = \frac{1}{2} - \frac{1}{2} \cos\left(\frac{6x}{\pi} + 4\right)$ Error $= 2.54 \times 10^{-5} < 5 \times 10^{-4}$

So we use $n=4$

$f^{(2n)}(x) \leq \frac{1}{2} \left(\frac{6}{\pi}\right)^{2n}$

$\int_1^1 f(x) dx = \sum_{i=0}^3 A_i f(x_i)$

$= 0.34719 \times \left(\frac{1}{2} - \frac{1}{2} \cos\left(\frac{6 \times 0.8611}{\pi} + 4\right) \right) + 0.34719 \times \left(\frac{1}{2} - \frac{1}{2} \cos\left(4 - \frac{6 \times 0.8611}{\pi}\right) \right)$

$+ 0.65281 \times \left(\frac{1}{2} - \frac{1}{2} \cos\left(\frac{6 \times 0.3399}{\pi} + 4\right) \right) + 0.65281 \times \left(\frac{1}{2} - \frac{1}{2} \cos\left(4 - \frac{6 \times 0.3399}{\pi}\right) \right)$

$\approx \underline{\underline{1.1095}}$

Question 4

here $a=0, b=0.1$

$$\text{Max error} \leq \frac{hM}{2L} [e^{L(b-a)} - 1]$$

$$y' = \frac{2 \cos x}{2+y^4} + \sin(1+x^8) = f(x, y)$$

$$|f(x, y_1) - f(x, y_2)| = |2 \cos x| \left| \frac{1}{2+y_1^4} - \frac{1}{2+y_2^4} \right|$$

$$\text{Assume } g(y) = \frac{1}{2+y^4} \quad |g(y_1) - g(y_2)| = g'(\xi) |y_1 - y_2|$$

$$\text{assume } \xi^4 = t, \xi^3 = t^{\frac{3}{4}} \quad g'(\xi) = -\frac{4\xi^3}{(2+\xi^4)^2} \quad h(t) = \frac{-4t^{\frac{3}{4}}}{(2+t)^2} \quad h'(t) = \frac{-3t^{-\frac{1}{4}}(2+t)^2 + 4t^{\frac{3}{4}}(2t+4)}{(2+t)^3}$$

$$\text{Take } t = \frac{6}{5}, \text{ then } |g'(\xi)| \leq \frac{4 \times (\frac{6}{5})^{\frac{3}{4}}}{(2+\frac{6}{5})^2} \approx 0.44 \quad |2 \cos x| \cdot |g'(\xi)| \leq 2 \times 0.44 < 0.9$$

we take $L = 0.9$

$$y' = f, \quad y'' = f_x \cdot \frac{\partial x}{\partial x} + f_y \cdot y' \quad |y''| \leq |f_x| + |f_y \cdot y'|$$

$$f_x = -\frac{2 \sin x}{2+y^4} + 8x^7 \cos(1+x^8) \quad |f_x| \leq \left| \frac{2 \sin x}{2+y^4} \right| + |8x^7 \cos(1+x^8)| \leq 1 + 8 \times 10^{-7}$$

$$|f_y| = \left| \frac{-2 \cos x \cdot 4y^3}{(2+y^4)^2} \right| \leq 0.9 \quad |y'| \leq \left| \frac{2 \cos x}{2+y^4} \right| + |\sin(1+x^8)| \leq 2$$

$$\therefore M \leq 1 + 8 \times 10^{-7} \rightarrow 2 \times 0.9 < 2.81, \text{ take } M = 2.81$$

$$\begin{aligned} \text{Error} &\leq \frac{hM}{2L} (e^{L(b-a)} - 1) \\ &= \frac{0.1 \times 2.81}{2 \times 0.9} (e^{0.9 \times 0.1} - 1) \\ &\approx 1.4 \times 10^{-4} < 0.0002. \quad \square \end{aligned}$$

Question 5

(a) Taylor's method: $Y_{i+1} = Y_i + h Y'_i + \frac{h^2}{2!} Y''_i$

$$y' = f = \frac{\cos(2xy)}{\sqrt{1+y^2}} \quad y'' = f_x + f_y \cdot y' \quad h = 0.1 \quad \text{here } x_0 = 0, y_0 = 1$$

$$f_y = \frac{-2x \sin(2xy) \sqrt{1+y^2} - y(1+y^2)^{-\frac{3}{2}} \cdot \cos(2xy)}{1+y^2} \quad f_x = \frac{-2y \sin(2xy)}{\sqrt{1+y^2}}$$

① Step 1 $Y'_0 = f(x_0, y_0) = \frac{\cos 0}{\sqrt{1+1}} = \frac{1}{\sqrt{2}} \approx 0.707$

$$Y''_0 = f_x + f_y \cdot y'$$

$$f_x(x_0, y_0) = -2 \cdot 0 = 0$$

$$f_y(x_0, y_0) = 0 - \frac{1 \cdot (2)^{\frac{3}{2}}}{2} \approx -0.3536$$

$$\therefore Y_1 = 1 + 0.1 \times 0.707 + \frac{0.01}{2} \times (0 - 0.3536 \times 0.707) \approx 1.069$$

② Step 2 $x_1 = 0.1, y_1 = 1.069$

$$Y_2 = y_1 + h f(0.1, 1.069) + \frac{h^2}{2} (f_x + f_y \cdot y')$$

$$f(0.1, 1.069) = \frac{\cos(0.2 + 0.2 \times 1.069)}{\sqrt{1+1.069^2}} \approx 0.668$$

$$f_x(0.1, 1.069) = \frac{-2 \times 1.069 \sin(0.2 \times 1.069)}{\sqrt{1+1.069^2}} \approx -0.399$$

$$f_y(0.1, 1.069) \approx -0.366$$

$$\begin{aligned} \therefore Y_2 &= 1.069 + 0.1 \times 0.668 + \frac{0.01}{2} \times (-0.399 \times -0.366) \\ &\approx 1.1330 \end{aligned}$$

Question 5 (cont'd.)

(b) Initialize $x_0 = 0$, $y_0 = 1$, $h = 0.1$

$$\text{Step 1 } k_1 = h f(x_0, y_0) = 0.1 \cdot \frac{0.4}{\sqrt{2}} \approx 0.0707$$

$$k_2 = h f(x_1, y_0 + h f(x_0, y_0)) = 0.1 \cdot f(0.1, 1.0707)$$

$$\approx 0.1 \cdot \frac{0.4(2 \times 0.1 \times 1.0707)}{\sqrt{1 + 1.0707^2}} \approx 0.0683$$

$$y_1 = y_{0.1} = 1 + \frac{k_1 + k_2}{2} \approx \underline{1.0695}$$

Step 2 $x_1 = 0.1$, $y_1 = 1.0695$

$$k_1 = 0.1 \cdot \frac{0.4(2 \times 0.1 \times 1.0695)}{\sqrt{1 + 1.0695^2}} \approx 0.0683$$

$$k_2 = 0.1 f(0.2, 1.0695 + 0.0683) = 0.1 \cdot \frac{0.4(0.4 \times 1.1378)}{\sqrt{1 + 1.1378^2}} \approx 0.066$$

$$y_2 = y_{0.2} = 1.0695 + \frac{0.0683 + 0.066}{2} \approx \underline{1.136}$$

$$\therefore y_{0.2} \approx 1.136$$

(c) Initialize $x_0 = 0$, $y_0 = 1$

for each $i = 0, \dots, n-1$, set $u_0 = y_0 + h f(x_0, y_0) = 1 + 0.1 \times \frac{1}{\sqrt{2}} \approx 1.0707$

assume $k = 0, 1, \dots, 4$

$$u_{k+1} = y_i + \frac{h}{2} [f(x_i, y_i) + f(x_{i+1}, u_k)]$$

① $k=0$

$$u_1 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, u_0)]$$

$$= 1 + 0.05 \left[\frac{1}{\sqrt{2}} + 0.6826 \right] \approx 1.0695$$

② $k=1$

$$u_2 = 1 + 0.05 \left[\frac{1}{\sqrt{2}} + f(0.1, 1.0695) \right]$$

$$\approx 1.0695$$

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Each step after this, the value of u_{k+1} changes a little bit

$$\therefore y(0.1) = u_5 \approx \underline{1.0695}$$

Question 6

for y' , assume $\frac{dy}{dx} = y' = g(x, y) = f^2(x, y)$

check if $g(x, y)$ has Lipschitz constant.

assume $f(x, y) = y$ $|f(x, y_1) - f(x, y_2)| \leq L |y_1 - y_2|$

we have Lipschitz constant $L=1$.

$$\geq \frac{dy}{dx} = y^2 \quad \int \frac{1}{y^2} dy = \int \frac{1}{y^2} dx$$

$$-y^{-1} = x + C \quad y = -\frac{1}{(x+C)}$$

Substitute $y(0) = 1$, $C = -1$

$$\therefore y = \frac{1}{1-x} \quad \text{But } y \text{ is on } x=1, \text{ no definition.}$$

So the statement is false